

Accessibility conditions of pronouns in *The Bounds of Meaning*

Abstract. This article is a commentary on the recent bound theory of anaphora in Mandelkern (2024), focusing on the conditions under which pronouns can be licensed in this theory (the *accessibility conditions*). There is a long-standing intuition that the accessibility conditions of pronouns with indefinite antecedents mirror the conditions on presupposition satisfaction, a parallel formalized in Hofmann (2022). Most theories do not fully explain Hofmann’s generalization (Chatain 2025). I argue that, in its basic form, the bound theory comes very close to validating the generalization and, furthermore, that in cases where it does not, Hofmann’s generalization itself appears to be incorrect, while the bound theory makes the right prediction. However, this article also finds that in other cases, the bound theory will require a notion of accommodation for pronouns. While a process of accommodation is known to exist for presuppositions, the analogous accommodation process that Mandelkern needs to posit for pronouns appears to need very strict limitations, which are hard to state and may not be needed in other theories.

Keywords. pronouns, anaphora, bounds, accessibility conditions.

I. Accessibility conditions for pronouns

When can pronouns such as *she*, *they*, *he* be used felicitously? This is the problem of *accessibility conditions*. For pronouns with indefinite antecedents (henceforth *discourse pronouns*), there is a long-standing intuition that the accessibility conditions mirror the conditions on presupposition satisfaction. More concretely, to every successful use of the pronoun in (1)a-d corresponds a felicitous use of the presupposition trigger *still* (which presupposes the past existence of a phone-book). Conversely, every infelicitous use of the pronoun in (1)e-g corresponds to an infelicitous use of *still* in (2)e-g.

- (1) Pronouns
 - a. There was a phone-book and *it* was in the cabinet.
 - b. If there was a phone-book, *it* was in the cabinet.
 - c. There might have been a phone-book and *it* would have been in the cabinet.
 - d. Either there wasn’t a phone-book or *it* was in the cabinet.
 - e. # There wasn’t a phone-book and *it* is in the cabinet.
 - f. # Either there was a phone-book or *it* was in the cabinet.
 - g. # There might have been a phone-book and *it* was in the cabinet.
- (2) Presuppositions
 - a. There was a phone-book and there *still* is.
 - b. If there was a phone-book, there *still* is.
 - c. There might have been a phone-book and there would *still* be one.
 - d. Either there wasn’t a phone-book or there *still* is.
 - e. # There wasn’t a phone-book and there *still* is.
 - f. # Either there was a phone-book or there *still* is.
 - g. # There might have been a phone-book and there *still* is.

Hofmann (2022), formalizing these long-standing intuitions (cf. also Mandelkern and Rothschild (2019) for a very similar formulation), proposed the following generalization, here presented in the formulation of Chatain (2025):

- (3) **Hofmann’s generalization:** If a discourse contains “[*a/some* RESTR] SCOPE” and a pronoun *he/she/it*, the pronoun may co-vary with the indefinite if and only if the presupposition that there exists an element in the intersection of the denotation of RESTR and the denotation of SCOPE can be felicitously satisfied in the position where the pronoun is used.

While this parallel between pronouns and presuppositions is well-recognized and often explicitly invoked as a motivation, [Chatain \(2025\)](#) argues that theories of anaphora, in particular in the Dynamic Semantics tradition, fall short of validating the generalization above in full. As a result, they may make mispredictions, for instance failing to predict the so-called bathroom cases in (1)d.

In his new book *The Bounds of Meaning*, Mandelkern makes an interesting new proposal for pronouns based on the new concept of *bounds*. This theory has some very desirable features. First, it leaves intact the truth-conditional semantics of classical first-order logic, adding to it only an extra independent layer of meaning (the bounds). Second, it has a wide empirical coverage. It successfully derives many of the cases which have proved difficult for previous approaches, including the paradigm above in (1). Third, Mandelkern’s bound theory can be motivated on independent grounds; he shows the usefulness of *bounds* in other empirical domains (such as modals and conditionals).

In this piece, my goal is to understand what accessibility conditions are predicted by Mandelkern’s bound theory, whether these accessibility conditions are in line with Hofmann’s generalization, and, if not, whether the predictions of Mandelkern are more or less correct than those made by Hofmann’s generalization. To do so, I will first propose a reformulation of Mandelkern’s bound theory which makes it easier to see the generalization about accessibility conditions (section II). I will argue that, while seemingly unified, the bounds of Mandelkern’s theory may cover two types of content with distinct pragmatic and projection behavior. Equipped with this dichotomy, I will, in section III, state accessibility conditions for Mandelkern’s bound theory and compare them to the accessibility conditions of theories based on Hofmann’s generalization, such as [Chatain \(2025\)](#). The conclusion will be mixed: in some cases, [Chatain \(2025\)](#) over-generates because it allows accommodation liberally; in others, Mandelkern under-generates and must resort to accommodation. Given the role of accommodation and the constraints on this process revealed by the previous discussion, section III will discuss whether accommodation and constraints on accommodation can help with some of the incorrect predictions in Mandelkern. I claim that accommodation in Mandelkern must be constrained in non-obvious ways or risk facing some of the same over-generation issues faced by [Chatain \(2025\)](#).

II. A reformulation of Mandelkern’s system

This section is an abridged presentation of Mandelkern’s approach to pronouns. The reader is invited to refer to [Mandelkern \(2024\)](#) for details. My goal with this presentation is to introduce my reformulation of his theory in terms of index and context bounds.

As discussed in the introduction, a major selling point of Mandelkern’s system is that it preserves both the classical translation of natural language sentences in first-order logic formulas and the truth-conditional semantics. So (4)a is indeed translated as (4)b and (5)a as (5)b.

- (4)
 - a. A woman came in and she sat down.
 - b. $[\exists x : \text{woman}(x), \text{came-in}(x)] \wedge \text{sat-down}(\iota x)$
- (5)
 - a. If a woman came in, she sat down.
 - b. $[\exists x : \text{woman}(x), \text{came-in}(x)] \rightarrow \text{sat-down}(\iota x)$

These formulas receive the traditional and unsurprising semantics for first-order logic formulas, as given in (6). Given a lexicon $\mathcal{F}$, formulas are evaluated for truth with respect to a world w and an assignment function g , a partial function mapping variables to

entities. The main difference¹ is that we annotate variables corresponding to pronouns with a ι ; this ι plays no role in the truth-conditional part of the theory.

(6)

- a. $q(x, (\iota)y, z, \dots)$ is true wrt (w, g) iff $\langle g_x, g_y, g_z, \dots \rangle \in \mathcal{J}_w(q)$
- b. $\neg p$ is true wrt (w, g) iff p is false wrt (w, g) .
- c. $p \wedge q$ is true wrt (w, g) iff p is true wrt (w, g) and q is true wrt (w, g) .
- d. $p \vee q$ is true wrt (w, g) iff p is true wrt (w, g) or q is true wrt (w, g) or both.
- e. $[\exists x : p, q]$ is true wrt (w, g) iff there is some g' differing from g only in the value assigned to x such that p is true wrt (w, g') and q is true wrt (w, g') .
- f. $[\forall x : p, q]$ is true wrt (w, g) iff for all g' differing from g only in the value assigned to x such that p is true wrt (w, g') , q is true wrt (w, g') .

In addition to truth conditions, Mandelkern proposes that formulas carry *bounds*. These bounds of a formula are relative to a triplet (c, w, g) , where c is a set of world-assignment pairs representing the context and (w, g) is a world-assignment pair, representing an index of evaluation. Both existentials and pronouns introduce bounds, as detailed in (7). When a formula has its bounds satisfied, we say that it is *satt*.

(7) Bounds

- a. $[\exists x : p, q]$ is *satt* at (c, w, g) only if: when $[\exists x : p, q]$ is true wrt (w, g) , $p \wedge q$ is true wrt (w, g) .
- b. ιx is *satt* at (c, w, g) only if for all $(w', g') \in c$, g'_x is defined.

The semantics is thus given as a combination of truth and bounds. In the pragmatics, truth and bounds jointly determine how the context is updated, as specified by the bridge principle in (8). This bridge principle is designed so that bounds, unlike presuppositions, need not be satisfied in the context prior to the update. Their content is automatically “added” into the context.

(8) Mandelkern’s bridge principle:

Upon successful assertion of a sentence S with meaning p , the context c is updated to $\{(w, g) \in c \mid p \text{ is true and satt at } (c, w, g)\} =: c^p$.

With this bridge principle, asserting the sentence in (9) will result in a context in which the assignment function g maps the variable x to a woman that came in, as in (10).

(9) A woman came in.

$\exists x : \text{woman}(x), \text{came-in}(x)$

(10)

$c \rightarrow c^{[\exists x : \text{woman}(x), \text{came-in}(x)]} = \{(w, g) \in c \mid g_x \in \mathcal{J}_w(\text{woman}) \cap \mathcal{J}_w(\text{came-in})\}$

Interpreted in the context created by (9), subsequent free occurrences of the variable x , as in (11), will correctly refer to some woman that came in. In addition to correctly referring, the pronoun in (11) will have its bound satisfied at any triplet $(c^{[\exists x : \text{woman}(x), \text{came-in}(x)]}, \cdot, \cdot)$ where the context coordinate is the output context in (10).

(11) She sat down.

$\text{sat-down}(\iota x)$

Bounds: $\forall (w, g) \in c^{[\exists x : \text{woman}(x), \text{came-in}(x)]}, g_x$ is defined

Now, consider (11) when uttered out of the blue. The out-of-the-blue context c_0 is modeled as a context containing all possible world-assignment pairs. In such a context,

1 Here, I deviate slightly from Mandelkern’s presentation. Mandelkern has a unified treatment of definites and pronouns and gives both the syntax of quantified expressions. Since definites play no role in what follows, pronouns are treated as variables, albeit special ones.

(11) is not satt for any (c_0, w, g) where c_0 is the first coordinate. By the bridge principle, the update will result in an empty context and thus presumably infelicity. This correctly predicts that pronouns cannot occur discourse-initially.

$$(12) \text{ Update of } c_0 \text{ by (11)} \\ c_0 \rightarrow c_0^{[\text{sat}(ix)]} = \{(w, g) \in c_0 \mid \forall (w', g') \in c_0, g'_x \text{ is defined}\} = \emptyset$$

The bridge principle we have discussed thus far describes the dynamics of context at the discourse level. At the sentence-internal level, Mandelkern defines projection rules for bounds, as in (13). The rule for conjunction in (13)b for instance states the first conjunct's bound will be evaluated with respect to the matrix context c and index (w, g) and that the second conjunct will be evaluated at the same index but in the context obtained by updating with the first conjunct. This in essence makes conjunctions equivalent to sentence sequencing.

- (13) Projection rules
- a. $\neg p$ is satt at (c, w, g) iff p is satt at (c, w, g) .
 - b. $p \wedge q$ is satt at (c, w, g) iff p is satt at (c, w, g) and q is satt at (c^p, w, g) .
 - c. $p \vee q$ is satt at (c, w, g) iff p is satt at $(c^{\neg q}, w, g)$ or q is satt at $(c^{\neg p}, w, g)$.

1. Index and context bounds

Although bounds depend in principle on both a context c and an index of evaluation (w, g) , the two expressions that introduce bounds in (7) only use one of these two dependencies: the bounds of the existential $[\exists x : p, q]$ depend only on (w, g) and the bounds of the variable ix depend only on c . Let us call bounds that only depend on (w, g) *index bounds* (i.e. the bounds of an indefinite) and let's call *context bounds* the bounds that only depend on c (i.e. the bounds of a pronoun).

Making this distinction helps us reformulate the pragmatics and projection of bounds, separating the case of index bounds from the case of context bounds. Consider a sentence with only a *context bound*, like (14). Since the context bounds only depend on the context c , Mandelkern's bridge principle in (8) entails that the updated context will either be the original c itself (if the bound is satisfied in c) or will be the empty set (if it isn't). The latter case is a failed update and will result in infelicity. This is what happens in (14) if x isn't valued in c . In other words, the felicity conditions of *context bounds* are like those of presuppositions: context bounds must be met throughout the context for the sentence to be felicitous.

- (14) She_x's happy
- (14) is satt in c iff for all $(w, g) \in c$, g_x is defined.
 - (14) is felicitous iff for all $(w, g) \in c$, g_x is defined.

Turning now to the projection rules, we may simplify the rules for the case of context bounds. Let's write $p^\bullet$ for a constituent containing a pronoun somewhere within it. It triggers a context bound which I call *pro*. Given the rules of projection in (13), *pro* will project in larger constituents as in (15). This follows from the projection rules and taking into account that *pro* only depends on c for its satisfaction.

- (15) Projection rules for context bounds
- a. *pro* is satisfied in $\neg p^\bullet$ at c iff *pro* is satisfied in $p^\bullet$ at c .
 - b. *pro* is satisfied in $p \wedge q^\bullet$ at c iff *pro* is satisfied in $p^\bullet$ at c^p .
 - c. *pro* is satisfied in $p^\bullet \wedge q$ at c iff *pro* is satisfied in $p^\bullet$ at c .
 - d. *pro* is satisfied in $p \vee q^\bullet$ at c iff *pro* is satisfied in $p^\bullet$ at $c^{\neg p}$.
 - e. *pro* is satisfied in $p^\bullet \vee q$ at c iff *pro* is satisfied in $p^\bullet$ at $c^{\neg q}$.

The rules derived in (15) are simply the rules for "Middle Kleene" projection, a system widely assumed to be adequate for presuppositions. Piecing these conclusions together,

we find that context bounds project like presuppositions and must, like presuppositions, be met throughout a context for a sentence to be felicitous.

Index bounds, on the other hand, have a different pragmatic and projection profile. On the pragmatic side, they are typically informative, i.e. they sieve out indices from the context and do not generally cause crashes, cf (16):

- (16) A woman_x came in.
 (16) is satt at (w, g) iff when $\mathcal{I}_w(\text{woman}) \cap \mathcal{I}_w(\text{came in}) \neq \emptyset$,
 $g_x \in \mathcal{I}_w(\text{woman}) \cap \mathcal{I}_w(\text{came in})$
 $c \rightarrow c^{\llbracket \exists x: \text{woman}(x), \text{came-in}(x) \rrbracket} = \{(w, g) \in c \mid g_x \in \mathcal{I}_w(\text{woman}) \cap \mathcal{I}_w(\text{came in})\}$

The projection rules of index bounds are straightforward. If $p^\bullet$ carries an index bound *indef*, it will simply project in larger constituents untouched cannot be filtered. In other words, they obey Weak Kleene projection rules.

- (17) Projection rules for index bounds
- a. *indef* is satisfied in $\neg p^\bullet$ at (w, g) iff *indef* is satisfied in $p^\bullet$ at (w, g) .
 - b. *indef* is satisfied in $p \wedge p^\bullet$ at (w, g) iff *indef* is satisfied in $p^\bullet$ at (w, g) .
 - c. *indef* is satisfied in $p^\bullet \wedge p$ at (w, g) iff *indef* is satisfied in $p^\bullet$ at (w, g) .
 - d. *indef* is satisfied in $p \vee p^\bullet$ at (w, g) iff *indef* is satisfied in $p^\bullet$ at (w, g) .
 - e. *indef* is satisfied in $p^\bullet \vee p$ at (w, g) iff *indef* is satisfied in $p^\bullet$ at (w, g) .

A possible analogue of this projection and pragmatic behavior can be found in supplements (Potts 2005). Supplements project from questions, negation and modals, as in (18). But they cannot be “filtered” like presuppositions, cf (19), and are not required to be part of the context prior to the update, as (20) shows.

- (18) Projection
- a. Did Mary talk to James, who is her cousin?
 $\rightarrow$ James is Mary’s cousin
 - b. Mary didn’t talk to James, who is her cousin.
 $\rightarrow$ James is Mary’s cousin
 - c. Mary may have talked to James, who is her cousin.
 $\rightarrow$ James is Mary’s cousin
- (19) # Either James, who is Mary’s cousin, is French or Mary and James aren’t cousins.
- (20) A: Mary talked to James, who is her cousin.
 B: #Wait a minute! Was I supposed to know that James and Mary are cousins?

Taking this line of reasoning one step further, one possible reinterpretation of Mandelkern’s theory is that the index carried by indefinites is a supplement, along the lines of (21). This is an intriguing possibility, which I will not pursue further.

- (21) A woman, which x is the name of, came in.

Table 1 summarizes our reformulation of Mandelkern’s theory

Table 1: Properties of index and context bounds

	Introduced by	Projection	Pragmatics	Closest equivalent
Index bounds	Pronouns	Weak Kleene	Are added to context	Presuppositions
Context bounds	Indefinites	Middle Kleene	Must be met in local context	Non-restrictive relative clauses

2. The treatment of attitudes in Mandelkern's bound theory

Since some of the cases below involve attitudes, I will briefly present Mandelkern's theory of pronouns under attitude verbs. To deal with anaphoric reference in attitude verbs, Mandelkern allows pronouns to refer to individual concepts (functions from worlds to individuals). The introduction of individual concepts is well motivated: indefinites under attitudes allow subsequent pronouns but only in certain configurations, cf (22). Individual concepts capture this by allowing a pronoun to have a value in some worlds (e.g. Mary's belief worlds) but not others (e.g. worlds of the common ground).

- (22) Mary believes that there is **a ghost**...
- a. ... She also believes **it** likes to make pranks.
 - b. ... # **It** likes to make pranks.

With this modification, g_x is an individual concept and the individual it picks out at world w is written as g_{xw} . The semantics must be amended as in (23).

$$(23) q(x, (t)y, z, \dots) \text{ is true wrt } (w, g) \text{ iff } \langle g_{xw}, g_{yw}, g_{zw}, \dots \rangle \in \mathcal{F}_w(q)$$

The context bound of a pronoun, in (24), is a straightforward extension of the original one: it requires all assignments in the context c to be defined for x and that the domain of the individual concept g_x includes all worlds in the context c .

$$(24) tx \text{ has its bounds satisfied at } (c, w, g) \text{ only if for all } (w', g') \in c, g'_{xw'} \text{ is defined.}$$

Likewise, the index bounds of indefinites do not change:

$$(25) [\exists x : p, q] \text{ has its bounds satisfied at } (c, w, g) \text{ only if: when } [\exists x : p, q] \text{ is true wrt } (w, g), p \wedge q \text{ is true wrt } (w, g).$$

Attitude verbs are introduced and, following the promise of Mandelkern's bound theory, they are given an entirely standard Hintikka semantics.

$$(26) B_a p \text{ is true at } (w, g) \text{ iff for all } w' \in B_{aw}, p \text{ is true at } (w', g) \\ \text{where } B_{aw} \text{ is the set of } a\text{'s doxastic alternatives at } w.$$

Attitude verbs project bounds, following the rules given in (27).

$$(27) B_a p \text{ is satt at } (c, w, g) \text{ iff, for every } \langle w', g' \rangle \text{ in } c \text{ and every } w'' \text{ in } B_{aw}, p \text{ is satt at } (B_{aw} \times \{g'\}, w'', g)$$

As before, we can separate out the rules for (27) into (28)a for context bounds and (28)b for index bounds. (28)a says that a context bound in the complement must be met in a certain modified context that contains all worlds accessible from the original context ; the assignments are obtained from the original context as well. The projection of index bounds in (28)b is noteworthy: unlike in the propositional fragment, index bounds do not project wholesale from attitude verbs ; rather, they are relativized to a 's doxastic alternatives.

$$(28) \\ \text{a. } pro \text{ is satt in } B_a p^\bullet \text{ at } c \text{ iff for every } \langle w', g' \rangle \text{ in } c, pro \text{ is satt in } p^\bullet \text{ at } B_{aw} \times \{g'\} \\ \text{b. } indef \text{ is satt in } B_a p^\bullet \text{ at } (w, g) \text{ iff for every } w'' \in B_{aw}, indef \text{ is satt in } p^\bullet \text{ at } (w'', g)$$

Let us illustrate these rules. In an out-of-the-blue context, (29) will trigger an update of the context that only keeps (w, g) in which Mary believes that there is a ghost in the world and where g maps x to said ghost in all of Mary's doxastic alternatives.

$$(29) \text{ Mary believes that there is } a_x \text{ ghost.}$$

$$B_m(\exists x : \text{ghost}(x), \text{there}(x)) \\ (30) c^{[B_m(\exists x : \text{ghost}(x), \text{there}(x))]} = \{(w, g) \mid \forall w' \in B_{aw}, g_{xw'} \in \mathcal{F}_{w'}(\text{ghost}) \cap \mathcal{F}_{w'}(\text{there})\}$$

This update creates a context suitable for a pronoun referring to the ghost, as in (31), so long as this pronoun is within the scope of an attitude verb ranging over Mary’s doxastic alternatives. More precisely, the pronoun introduces the context bound in (32), which is met in the context c_0 defined in (30).

(31) She also believes it_x likes to make pranks.

$B_m(\text{pranks}(x))$

(32) (31) is satt in c_0 iff for every $\langle w', g \rangle$ in c and for every w'' in $B_{aw'}$, $g_{xw''}$ is defined

3. Taking stock

This reformulation of Mandelkern’s system in terms of context and index bounds serves two purposes. First and foremost, it is meant to simplify the upcoming discussion. Second, it highlights a theoretical concern: while unified in the formalism, Mandelkern’s context and index bounds, have little in common ; they differ in both their projection and pragmatic behavior. This suggests an alternative formalization where index bounds are treated as run-of-the-mill presuppositions and context bounds constitute the new category of content posited by Mandelkern. I will not attempt this reformalization here but point to Spector (2024), who builds on Mandelkern in a way entirely parallel to what is suggested here.

III. Mandelkern and the existence generalization

1. Accessibility conditions in Mandelkern (2024)

With the reformulation from the previous section, we are ready to evaluate what accessibility conditions Mandelkern’s bound theory predicts for pronouns and how it compares to Hofmann’s generalization, repeated in (33).

(33) **Hofmann’s generalization:** If a discourse contains “[*a/some* RESTR] SCOPE” and a pronoun *he/she/it*, the pronoun may co-vary with the indefinite if and only if the presupposition that there exists an element in the intersection of the denotation of RESTR and the denotation of SCOPE can be felicitously satisfied in the position where the pronoun is used.

It is useful to start by seeing how a theory based on Hofmann’s generalization captures the patterns of felicity and infelicity in (34).

(34) Pronouns

- a. There was a phone-book and it was in the cabinet.
- b. If there was a phone-book, it was in the cabinet.
- c. There might have been a phone-book and it would have been in the cabinet.
- d. Either there wasn’t a phone-book or it was in the cabinet.
- e. # It was in the cabinet and there was a phone-book.
- f. # There wasn’t a phone-book and it was in the cabinet.
- g. # Either there was a phone-book or it was in the cabinet.

As a baseline, I will take my own proposal in Chatain (2025), hereafter Chatain. In a nutshell, the proposal takes (33) at face value and assumes that the (discourse) pronoun presupposes the existence of an element in the restrictor and scope of its indefinite (i.e. a witness). More precisely, this theory assumes that indices borne by pronouns and indefinites name properties. An indefinite DP may only be indexed i if i names the set of its witnesses (the property $\lambda x. \text{phone-book}(x) \wedge \text{there}(x)$ in (35)). A pronoun presupposes that the property named by its index is non-empty and picks its referent among the extension of the property (via a choice function)

(35) There was a phone-book_i and it_i was in the cabinet.

(36) $\llbracket \text{it}_i \rrbracket = [G_i \neq \emptyset] f(G_i)$

where G_i is the property named by i and f a choice function.

The technical details won't matter much to us here. The important feature of this account is that pronouns carry a presupposition that the set of witnesses for its antecedent is not empty. In the case of (35), "it_i" presupposes that there are phone-books there and this presupposition can be trivially met by the first conjunct.

To apply this theory to the larger dataset in (34), let us assume that all sentences occur in a discourse with no preceding indefinite (which would affect the judgments given below). In all sentences of (34), Chatain predicts the pronoun will be acceptable only on the condition that the presupposition "there is a phone-book" can be met at the point where the pronoun is used. In (34)a to (34)d, following standard theories of presupposition, this presupposition can be filtered by the material preceding the pronoun. The sentences as a whole carry no presupposition and can therefore be felicitous in any context. The generalization correctly expects these sentences to be felicitous.

The infelicity of sentences (34)e to (34)g requires a more involved explanation. In (34)e, the presupposition "there is a phone-book" cannot be filtered in the position where the pronoun is used. So the only way the presupposition can be met is if it is already entailed by the context or accommodated into it. But contexts which entail that there is a phone-book are contexts in which it is redundant to assert "and there is a phone-book". So (34)e is infelicitous either way: either the context cannot meet the pronoun's presupposition or it makes the utterance itself redundant. Similar considerations explain the infelicity of sentences (34)f to (34)g. In (34)f for instance, the presupposition that there is a phone-book can only be met in contexts where the utterance would be contradictory. In (34)g, the same presupposition makes the disjunction equivalent to its second disjunct, a violation of Manner.

Mandelkern's explanation of the felicitous cases in (34)a to (34)d is very similar to Chatain's: the pronoun in these cases triggers a context bound that x must be valued; this context bound can be 'filtered' by the preceding indefinite ; context bounds, after all, project and are filtered according to the same rules as presuppositions. The infelicitous cases (34)e to (34)g receive a somewhat different explanation. In these cases, the pronoun's context bound cannot be filtered by the indefinite ; it must therefore be met in the global context. Meeting this context bound requires there to have been some indefinite in previous discourse valuing x . But we assumed earlier that this was not the case and thus, the sentence must be infelicitous.

(37) Context bounds for sentences (34)e to (34)g

a. g_x is defined throughout c

b. g_x is defined throughout $c^{[\neg \exists x, \text{phone-book}(x)]}$

c. g_x is defined throughout $c^{[\neg \exists x, \text{phone-book}(x)]}$

This exploration highlights a critical difference between Mandelkern's system and Chatain's. In Chatain, the pronoun's presupposition must be met in some way or other. One way of meeting the presupposition is through filtering by the indefinite's antecedent. But that need not be the case: the presupposition could be met by accommodation or simply because the context already entails it. This means that explaining infelicity in Chatain's theory requires an extra step: the theory must explain why the presupposition could not be met in one of these other ways. Explanations vary from example to example: ban on redundant conjuncts, unnecessary disjuncts, maxim of Manner, etc. In Mandelkern's system (as presented thus far), the explanation is more straightforward: filtering by an indefinite antecedent is the only way of licensing a pronoun. No extra step is required to explain infelicity.

More generally, the accessibility conditions predicted by both theories are as in Table 2 below.

Table 2: Generalizations about accessibility conditions

Mandelkern	Hofmann
1. An indefinite must be present in the surrounding discourse.	1. An indefinite must be present in the surrounding discourse.
2. The local context of the pronoun entails the existence of a witness.	2. The local context of the pronoun entails the existence of a witness.
3. The indefinite antecedent filters the pronoun's bounds	

From a theoretical perspective, these explanations are matched in complexity. The extra step of Chatain's might look superfluous when compared to Mandelkern's approach but it must be remembered that this step is also needed to explain the infelicity of the presupposition triggers in (38) and thus independently motivated. On the flip side, the introduction of bounds in Mandelkern might look superfluous given that Chatain's treatment only requires a very standard theory of presupposition. But, again, this argument must be weighed against the independent motivations for bounds that Mandelkern provides (modals, conditionals, etc).

(38)

- a. # There wasn't a phone-book and there **still** is.
- b. # Either there was a phone-book or there **still** is.
- c. # There might have been a phone-book and there **still** is.

From an empirical perspective, the accessibility conditions predicted by the two theories are not equivalent and so we might expect some differences in predictions. This is what the subsequent discussion turns on.

2. Differences in empirical predictions.

We will present two cases differentiating Mandelkern's accessibility conditions from Chatain's. One will favor the former approach and the other the latter. Consider (39) as a starting point. In Mandelkern's theory, the pronoun's context bound in (39) cannot be filtered by a subsequent indefinite and so the sentence is predicted to be infelicitous. In Chatain, by contrast, the pronoun, while it requires an antecedent, does not need to have its presupposition filtered by this antecedent. In particular, it can be filtered by the first conjunct *James is a parent*. The pronoun is therefore licensed.

(39) # James is a parent, **she's** at boarding school and James has **a child**.

The only reason (39) is infelicitous in Chatain's theory is because the first conjunct *James is a parent* is redundant with the third *James has a child*. That redundancy explains the infelicity of (39) and can already be seen in the simpler (40).

(40) # James is a parent and James has a child.

If redundancy is the reason for (39)'s infelicity, then changing the contribution of the clause containing the indefinite so that it is no longer entailed by the first should improve (39). (41) provides an example: the second conjunct *she's at boarding school* containing the pronoun is not entailed by the third conjunct *many would be surprised to ...*; yet, the cataphoric pronoun remains infelicitous. The baseline in (42), without the pronoun, serves as a proof that our manipulation has successfully removed any redundancy.

- (41) # James is a parent, [she](#)'s at boarding school and many would be surprised to hear that James has a child.
- (42) James is a parent and many would be surprised to hear that James has a child.

So Chatain's accessibility conditions appear too loose. There are, however, cases where Mandelkern's accessibility conditions are too strict. These cases will require introducing a notion of 'bound accommodation' in Mandelkern's theory. With accommodation, Mandelkern's accessibility conditions become slightly looser and so, as we will see, the conclusion that the pronoun in (41) is infelicitous may no longer hold.

IV. Accommodation

1. The need for accommodation

Chatain (2025) presents cases where the indefinite isn't in a position to filter the pronoun's presupposition and yet the pronoun is possible. In (43), the pronoun can successfully co-vary with an indefinite antecedent nested under both negation and an attitude verb.

- (43) Mary believes I don't have [a child](#). And yet, she's seen [her](#) many times.

The interest of this example is that no theory of presupposition expects the first sentence to be able to filter the presupposition of the second sentence. In Chatain, this is not an issue since there is no requirement that the presupposition of the pronoun be filtered by the indefinite; it simply needs to be met. Here, the presupposition in question (that I have a child) is easily accommodated. Accommodation is furthermore facilitated by the fact that the sentence leads to an anti-factive inference that Mary's belief is false (Chemla 2008).

In Mandelkern, filtering (of context bounds) must take place for the pronoun to be licensed. But the indefinite's bound, being under negation, is vacuous given the assertion and therefore does not place any constraint on the assignment functions in the context. The pronoun's bounds cannot be 'filtered'.

- (44) Mary believes I don't have [a child](#)...

$$c^{[B_m(\neg\exists x:\text{child}(x),\text{mine}(x))]} = \left\{ (w,g) \left| \begin{array}{l} \forall w' \in B_{aw'}, \neg\exists x, x \in \mathcal{F}_{w'}(\text{child}) \cap \mathcal{F}_{w'}(\text{mine}) \\ \forall w' \in B_{aw'}, (\exists x, x \in \mathcal{F}_{w'}(\text{child}) \cap \mathcal{F}_{w'}(\text{mine})) \\ \rightarrow g_{xw'} \in \mathcal{F}_{w'}(\text{child}) \cap \mathcal{F}_{w'}(\text{mine}) \end{array} \right. \right\}$$

$$c^{[B_m(\neg\exists x:\text{child}(x),\text{mine}(x))]} = \{(w,g) \mid \forall w' \in B_{aw'}, \neg\exists x, x \in \mathcal{F}_{w'}(\text{child}) \cap \mathcal{F}_{w'}(\text{mine})\}$$

(The index bound introduced by the indefinite is vacuous under negation: whenever the existential is true, its restriction and scope must be true, but the negation requires the existential to be false. Thus the update imposes no additional condition on g .)

More generally, some examples of anaphora are supported by some amount of pragmatic inference. Nothing in the literal meaning of (45) entails that James has a secret child. But the cessation implicature from 'I long thought that p ' to 'not p ' is sufficient² to license the pronoun.

- (45) Alfred has long held the belief that James has [a secret child](#). I long thought it couldn't be true. But then, I saw James playing with [her](#) at the park.

2 In 10.1.2, Mandelkern argues that anaphora will be supported from any attitude as long as the attitude is contextually veridical, not just lexically veridical. My examples here are making the same point. However, as the discussion here shows, I disagree with his statement that "*the reasoning [for these cases] is just as for lexical factives*", because the cases here require accommodation, whereas lexical factives do not.

A similar problem is discussed by Mandelkern in section 10.1.2 in connection to the famous Hob-Nob sentences (Geach 1967): (46)a only guarantees that g_{xw} is defined for every world w of Hob's doxastic alternatives ; it does not ensure that g_{xw} is defined when w is one of Nob's doxastic alternatives. The pronoun in (46)b is not predicted.

- (46)
- a. Hob thinks a_x witch blighted Bob's mare ...
 - b. ... Nob thinks that she killed Cob's sow.

In response to (46), Mandelkern proposes a mechanism of “bound accommodation”, similar to the accommodation that Chatain appeals to. But this is more problematic than it appears. First, this is a fine line to walk: by loosening the accessibility conditions of his theory, Mandelkern risks facing the over-generation problems that affect Chatain's, which we discussed in section III.2. Second, a mechanism of “bound accommodation” is not as straightforward to formulate as standard presupposition accommodation. In the sequel, I propose to compare accommodation in the theories of Chatain and Mandelkern.

2. Formulating bound accommodation

The accommodation procedure for presuppositions is easy to define: when hearing a sentence whose denotation is undefined at some index of the context, the hearer may choose to remove from the context any index for which that proposition is undefined.

- (47) **Presupposition Accommodation:** If p is undefined for some $(w, g) \in c$, then update c to $c^{[Ap]}$ where $c^{[Ap]} = \{(w, g) \in c \mid p \text{ is defined at } (w, g)\}$

Stating (47) is easy because, under most analyses, a sentence's presuppositions are met or not met at an index: accommodation simply corresponds to an intersection with the indices where the presupposition is met.

The context bounds of pronouns, by contrast, depend on the whole context c . So we cannot directly write a rule of accommodation like (47) because there is no corresponding notion of a context bound being met *at an index*. However, we can note that all context bounds seen so far have had the following property: the bound was met at c if and only if it was met at all singleton contexts $\{(w, g)\} \subset c$. This is easy to check on the two representative examples in (48) and (49).

- (48) ... And yet, Mary's seen her_x many times
Context bounds: *for all* (w, g) in c , g_{wx} is defined
- (49) ... Nob thinks that she_x killed Cob's sow.
Context bounds: *for all* (w', g) in $\{(w', g) \mid \exists w, w' \in B_{\text{Nob}, w} \wedge (w, g) \in c\}$, $g_{w'x}$ is defined
Equivalently: *for all* (w, g) in c , *for all* $w' \in B_{\text{Nob}, w}$, $g_{w'x}$ is defined.

Given this, we *can* write an accommodation procedure for context bounds on the model of (47) after all:

- (50) **Context Bound Accommodation:** If p is not satt at c , then update c to $c^{[Ap]}$ where $c^{[Ap]} = \{(w, g) \in c \mid p \text{ is satt at } \{(w, g)\}\}$

But this accommodation rule turns out to be catastrophic for most intents and purposes, since it never places any significant constraints on the values that pronouns receive. For instance, the accommodation of the pronoun's context bound in (52) in the context created by (51) results in the extremely weak constraint that g_{xw} is defined. There is no requirement that *her* refer to my child.

- (51) Mary believes I don't have a child...
- After update:
 $c' = c^{[B_m(\neg \exists x: \text{child}(x), \text{mine}(x))]} = \{(w, g) \mid \forall w' \in B_{aw}, \neg \exists x, x \in \mathcal{J}_{w'}(\text{child}) \cap \mathcal{J}_w(\text{mine})\}$

(52) ... And yet, she's seen **her** many times.

After accommodation:

$$c'[A(\text{seen}(ix))] = \{(w, g) \mid g_{xw} \text{ is defined} \wedge \forall w' \in B_{aw}, \neg \exists x, x \in \mathcal{F}_w(\text{child}) \cap \mathcal{F}_w(\text{mine})\}$$

The broader point is that the core requirement of a pronoun in Mandelkern is simply that a certain index be defined. When accommodation is not used, this requirement can only be met under stronger conditions: there must be an indefinite and that indefinite must filter the pronoun's bounds. When accommodation is possible and filtering is not necessary, only the core requirement of the pronoun, which is very weak, remains. This contrasts with Chatain's theory ; there, the pronoun itself carries a strong requirement that its indefinite antecedent has a witness (e.g. that I have a child in (52)). This is so regardless of accommodation.

One could reasonably object that the accommodation proposed above is too mechanical. Accommodation is a pragmatic process that leaves room for speaker intention: hearers do not always mechanically accommodate only the weakest information needed for interpretation; they reason to what kind of context the speaker might have wanted to jump to. This flexibility in accommodation is arguably observed already in the case of presupposition accommodation. For example, the sentence in (53) carries a weak presupposition, but listeners may interpret the speaker as intending any of the inferences in (54) to be added to the common ground (a.k.a. the *proviso problem*, Geurts (1996); Beaver (2001); Mandelkern (2016)).

(53) If John is a Martian, his antennas must be buzzing right now.

presupposes: *if John is a Martian, he has antennas.*

(54) Possible accommodations:

a. *John has antennas.*

b. *Martians have antennas.*

c. *Male Martians have antennas.*

In this light, we could propose the more liberal accommodation rule in (55). This rule leaves it unspecified which context c' hearers should accommodate and relies on pragmatic reasoning to fill in the details.

(55) Context Bound Accommodation: If p is not satt at c , then update c to $c' \subset c$ such that the context bounds of p are met at c' .

This theory can explain the observed interpretation of the sequence in (51) and (52): there is indeed some context c'' that delivers the attested reading.

(56) ... And yet, she's seen **her** many times.

After accommodation:

$$c'' = \{(w, g) \mid \begin{array}{l} g_{xw} \in \mathcal{F}_w(\text{child}) \cap \mathcal{F}_w(\text{mine}) \\ \wedge \forall w' \in B_{aw}, \neg \exists x, x \in \mathcal{F}_w(\text{child}) \cap \mathcal{F}_w(\text{mine}) \end{array} \} \subset c'$$

The difficulty is of course to rule out many alternative accommodations which could give rise to unattested readings; (57) for instance would deliver a reading where *she* refers to Mary's child.

$$(57) c'' = \{(w, g) \mid \begin{array}{l} g_{xw} \in \mathcal{F}_w(\text{child}) \cap \mathcal{F}_w(\text{Mary's}) \\ \wedge \forall w' \in B_{aw}, \neg \exists x, x \in \mathcal{F}_w(\text{child}) \cap \mathcal{F}_w(\text{mine}) \end{array} \} \subset c'$$

Ruling out these unwanted readings isn't necessarily a hard task for (56): arguably, only a reading where *her* refers to my child forms a coherent discourse (especially given the speaker's choice of the contrastive *and yet*).

An important consequence of this accommodation account is that, when accommodation is used, the presence of an indefinite antecedent is no longer required to license a pronoun. This seems problematic: the contrast in (58) (adapted from a similar minimal pair in Mandelkern) shows that the presence of an indefinite does matter to the licensing of the

pronoun, even in these accommodation cases. To account for this contrast, Mandelkern suggests that the result of accommodation is dependent not just on the context bound that one is trying to meet but on available material in surrounding discourse (i.e. *have a child* in (58)a).

(58)

- a. Mary believes I don't have a_x child. And yet, she_x's seen her multiple times.
- b. Mary believes I am not a parent. # And yet, she's seen her multiple times.

In Chatain³, by contrast, the presence of the indefinite plays a role in determining what presupposition is carried by the pronoun. In (58)a, the co-indexing imposes that the pronoun presupposes that the speaker has a child and this is what *must* be accommodated. For lack of a potential antecedent in (58)b, the pronoun cannot be determined to carry a similar presupposition and thus the sentence is infelicitous.

But restricting accommodation to material overtly provided alone might not be sufficient. Recall from section III.2 that Chatain's theory struggles to explain the infelicitous pronoun in (59)a. Augmented with accommodation, Mandelkern's theory faces a similar challenge: accommodation can license the pronoun because there is suitable material in the surrounding discourse (the later clause *James has a child*). So an additional constraint on accommodation must be added if one wants to rule (59)a out. Perhaps, accommodation cannot rely on upcoming material, but only on material from the preceding discourse.

(59)

- a. # James is a parent, she's at boarding school and many would be surprised to hear that James has a child.
- b. James has a child, she's at boarding school and many would be surprised to hear that James has a child.

All in all, the weakness of the pronoun's requirements in Mandelkern's theory means that the accommodation procedure must be constrained in non-obvious ways and is liable to over-generate. Chatain's theory over-generates too but the pronouns carry more constraints, which limits what must be accommodated.

There are cases, however, where the pragmatic leeway in Mandelkern's theory seems beneficial: the Hob-Nob cases. From Hob-Nob sentences as in (60), we routinely infer that Nob has certain beliefs beyond those he is explicitly asserted to have. In particular, as Mandelkern suggests, it is natural to infer that Nob also believes that there's a witch running around. Such inferences, in Mandelkern, could be the result of accommodation. Importantly, it is usually under-specified which shared beliefs must be accommodated in Hob-Nob cases. Instead of (60)b, it is easy to infer that Nob holds a stronger belief, for instance that Nob also thinks that a witch blighted Bob's mare as in (60)c. Or, more marginally, a weaker belief, as in (60)a, where Nob doesn't believe in witches but does believe some woman with a questionable sense of fashion is running around causing troubles in the neighborhood. This variability in interpretation is natural if we think that the licensing of the pronoun requires a form of accommodation that is under-specified: in different contexts, accommodation of the same context bound may result in different information being accommodated.

(60) Hob thinks a_x witch blighted Bob's mare.

Nob thinks that she_x killed Cob's sow.

- a. possible accommodation: *Nob believes a woman with a funny hat is wreaking havoc in the neighborhood.*

3 Chatain recognizes a difficulty with non-co-indexed pronouns. Nothing in the grammar rules them out and so the pronouns could in principle carry any presupposition and any reference.

- b. possible accommodation: *Nob believes a witch is running around.*
- c. possible accommodation: *Nob believes a witch blighted Bob's mare.*

This seems like an advantage over Chatain. In Chatain, the pronoun is licensed only if the presupposition that a witch blighted Bob's mare is met in the pronoun's local context. This requires, at the matrix level, accommodating that Nob believes a witch blighted Bob's mare, i.e. (60)c. This is a possible reading (although see later) but it isn't clear *prima facie* how the theory could predict any of the weaker accommodations in (60).

3. Hob-Nob revisited

On closer inspection though, it is not clear that the variability in interpretation in the Hob-Nob case in (60) is the product of accommodation. It may instead reflect a more general underdetermination concerning how the objects of different attitude states are represented by their holders. As is well-known, a report containing a referential expression in the scope of an attitude verb can leave under-determined the way in which the attitude holder identifies or represents its apparent referent. Take (61), for instance. Nob need not share the speaker's beliefs about Samantha, nor need he think of her under the description *the witch who blighted Bob's mare*. Depending on the context, any of the beliefs in (62) might characterize the way in which Nob represents the person whom the speaker calls Samantha.

- (61) Let me call the witch that blighted Bob's mare Samantha. What I am saying is that Nob believes that Samantha killed Cob's sow.
- (62) Possible beliefs of Nob
 - a. *Some person is causing trouble around these parts.*
 - b. *There's a witch running around causing troubles.*
 - c. *A witch blighted Bob's mare.*

Could this independently motivated under-determination of the interpretation of DPs under attitudes also be responsible for some of the apparent under-determination of Hob-Nob sentences, rather than accommodation of an index?

One fact militates in favour of the affirmative. The central intuition associated with Hob-Nob sentences is not that Hob and Nob have similar beliefs about witches. Rather, the intuition present in the original Geach (1967), who introduced the example, is that their belief must have a "common focus". Very loosely speaking, they must be about "the same witch" (this is strictly speaking incorrect because no witch in fact exists). The "common focus" must be grounded either in some interaction (Hob and Nob have talked about it) or drawn from a shared evidential source (they have read about the putative witch in the local newspaper) or something else. Hob and Nob cannot simply have to come at these beliefs on completely independent grounds.

This intuition can be converted into a sharp judgment, as in (63), adapted from Edelberg (1986). (63) feels for instance inappropriate in the context described. It is not that the anaphora is infelicitous in this sentence; rather, its use seems to imply a connection between the beliefs of Hob and Nob that is absent in the context.

- (63) Context: *in village A, false rumours have been spreading that the perfectly natural death of some livestock was the act of a witch. In village B, quite independently, newspapers have been spreading around the rumour that some witch lives in the abandoned castle. Villages A and B are on opposite ends of the globe and unaware of each other.*
- # Hob, in village A, believes there is a witch running around and Nob, in village B, believes she lives in the local castle.

The puzzle of intentional identity raised by Geach is to understand how the beliefs of two individuals can in some sense have a common focus, without there being an actual object that both attitude holders are referring to. There is a good amount of literature (Saarinen

1978; Edelberg 1986; Edelberg 1992; van Rooy 2000; Salmon 2005; Cumming 2014; Ninan 2024) on this problem with different solutions which I will not review here. What is important for present purposes is the point that mere similarity in the contents of the two beliefs is not sufficient and some sort of identity, however stated, is involved.

The problem is that this identity condition does not follow from the accommodation account. In the accommodation account, the pronoun in the second conjunct is licensed by accommodating certain constraints on Nob's belief state, whatever information is required to make the index introduced under Hob's attitude accessible there. If what must be accommodated is merely that Nob believes that there is a witch, the theory will wrongly predict (63) to be felicitous and true, contrary to fact.

This shows two things: that the accommodation account is in fact, contra appearances, too liberal because it fails to explain the deviance of the Hob-Nob variation in (63). So the rule of accommodation must be further restricted so that it does not license a pronoun in (63). Second, it shows that there must be some alternative route to Hob-Nob cases that doesn't go through accommodation of indices. The important negative conclusion from this discussion is that accommodation, if it exists, must in fact be ruled out in Hob-Nob cases. If it were allowed, the sentence might receive a reading that wouldn't require 'common focus'. Since no such reading is attested, accommodation must not be possible here. It isn't clear why it wouldn't be possible. A similar issue arises for Chatain. In Chatain's theory, the pronoun in Hob-Nob cases will be valid as long as the presupposition that 'a witch blighted Bob's mare' can be accommodated in the local context. This is equivalent to a global presupposition that Nob believes that a witch blighted Bob's mare. The account therefore predicts that, as long as this presupposition can be accommodated, the sentence is predicted felicitous. But, if the reasoning above is correct, that presupposition on its own is insufficient to license the pronoun.

V. Summary

This commentary on Mandelkern's theory argues for three conclusions:

- First, that the "bounds" (at least as far as anaphora is concerned) are not unified formally, dividing themselves between the presupposition-like context bounds and the supplement-like index bounds.
- Second, that Mandelkern's theory predicts more stringent accessibility conditions than are predicted by Hofmann's generalization. This appears to be a desirable result, as Chatain, which is based on the latter generalization, faces over-generation issues.
- Third, that it is harder to state a process of "pronoun" accommodation in the bound theory than it is in a presuppositional theory à la Chatain. In the bound theory, the process relies on the hearer reconstructing the needed context bound entirely from context with little clue from the form itself. Despite the *prima facie* appearances from the Hob-Nob cases that this leeway may be desirable, "pronoun" accommodation appear in fact quite constrained in a way that all theories miss.

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